



VISVESVARAYA TECHNOLOGICAL UNIVERSITY

“Jnana Sangama”, Belagavi - 590018

MAJOR PROJECT PHASE – II PRESENTATION

“Autonomous drone for disease detection and pesticide spraying In Arecanut plantations”

Under the Guidance of:

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INTRODUCTION

- Arecanut farming in Karnataka face serious crop damage due to Koleroga disease (Fruit rot), especially during monsoon conditions.
- Manual pesticide spraying is difficult, labor intensive and unsafe due to tall and dense plantations. It also results in uneven spraying of the chemicals.
- The project proposes the development of an autonomous Drone based system which uses AI based disease detection.
- The drone is capable of performing targeted pesticide spraying, reducing chemical wastage and improving safety.
- The drone uses GPS based navigation to enable localization and autonomous movement across the plantation system.

LITERATURE SURVEY- KEY OUTCOMES

Sl.no	Paper title	Author	Year	Summary
1	Design and implementation of agricultural drone for Arecanut farms	Raju Hajare, C P. Mallikarjuna Gowda, and M. V. Sanjaya	2021	Details the development of a quadcopter with a Pixhawk Flight Controller for semi-autonomous fungicide spraying. It includes live video transmission and allows for circular trajectory tracing around trees.
2	Remote Sensing Detecting of Yellow Leaf Disease of Arecanut Based on UAV Multisource Sensors	Lei, Shuhan; Luo, Jianbiao; Tao, Xiaojun; Qiu, Zixuan	2021	Focuses on quantifying the severity of Yellow Leaf Disease (YLD) using multispectral data (NDVI, LCI, etc.). Utilizes BPNN and SVM algorithms for high-resolution classification and assessment of leaf volume loss.
3	Artificial Intelligence based drone for early disease detection and precision pesticide management in cashew farming	Manoj Kumar Rajagopal, Bala Murugan MS	2023	Employs an edge computing paradigm with MobileNetV2 to classify anthracnose disease in cashew leaves. Achieved 95% accuracy for disease identification and 99% for healthy leaves using drone-captured images.
4	UAV Platform for Pesticide Spraying and Disease Detection for Areca Nut and Pepper Plantations	Shrinidhi I S, Pratham K B, Raynol Menezes, Rajmohan Kamath, Ajith B S, Chandra Singh	2023	Features the OrcBOT autonomous orchard robot. It uses YOLOv5 for real-time canopy detection and geolocalization metadata to identify the precise geographical location of detected palm trees.
5	Plant disease detection using drones in precision agriculture	Ruben Chin , Cagatay Catal, Ayalew Kassahun	2023	A systematic literature review (SLR) identifying quadcopters as the most used drone type and CNNs as the primary ML task. It notes that fungus is the most common pathogen studied in this field.

Sl.no	Paper title	Author	Year	Summary
6	UAV Automatic Spraying System Based on Image Recognition of pests and diseases	Yongyi Yi, Chunling Lang, Chuanxin Xin, Haoxin Lan, Xiulong Gao, Guojian Lin, and Zhaoran Lian.	2023	Investigates the use of UAV swarms for monitoring larger areas. It focuses on using AI and ML to identify diseases and pests in Brassica plants, optimizing profitability through autonomous spraying.
7	A Review on Real-Time Computer Vision Based Robotic Pesticide Sprayer for Multi-Crop	Mr. Nagaraja Hebbar N Dr. Sandeep Bhat	2024	Reviews AI-powered systems that leverage CNN-based multi-class classification to identify diseases from leaf images. It emphasizes the integration of robotics for targeted, real-time pesticide application.
8	Deep Learning-Based Arecanut Disease Detection and Smart Cluster Management with Automated Pesticide Spraying System	Ashritha K P, Navaneeth Bhaskar, Lakshmi H N, Shaik Jaffar Hussain, Akepati Sankar Reddy, and B Rupa Devi.	2025	Integrates YOLOv8 for disease detection (96.2% accuracy) and k-Means clustering to segment arecanut bunches. Uses a Deep Q-Network (DQN) to optimize drone spray trajectories, reducing chemical waste by 30%.
9	Deep learning and computer vision in plant disease detection: a comprehensive review	Abhishek Upadhyay, Narendra Singh Chandel, Krishna Pratap Singh, Subir Kumar Chakraborty, Balaji M. Nandede, Mohit Kumar, A. Subeesh, Konga Upendar, Ali Salem & Ahmed Elbeltagi	2025	A comprehensive review analyzing over 278 research articles. It highlights state-of-the-art Deep Learning (DL) models, reference datasets, and recent trends in non-destructive image analysis for precision agriculture.
10	A Review on the Detection of Plant Disease Using Machine Learning and Deep Learning Approaches	Thandiwe Nyawose, Rito Clifford Maswanganyi, and Philani Khumalo.	2025	Critically evaluates model performance (like EfficientNetV2-S and Vision Transformers) in real-time field conditions. It discusses multimodal learning and edge deployment suitability for complex environments.

Problem statement

Arecanut is a commercial crop for many farmers, but its yield is reduced due to the disease Koleroga (Fruit Rot). Manual pesticide spraying for this disease is very time consuming, requires a large amount of labor, and is risky due to the tall height of trees. Due to this, there is significant crop loss, uneven distribution of chemicals and rise in health risks for farmers.

Therefore, to overcome the problems of manual plantation, monitoring and spraying, an autonomous drone system is used to reduce human risk and avoid manual climbing. The system uses ROS for smooth communication between navigation and control. Along with this, it also uses GPS based localization for autonomous movement across plantation rows. Machine learning models ensure real time disease detection, while a targeted spraying system carries out uniform pesticide distribution with reduced wastage.

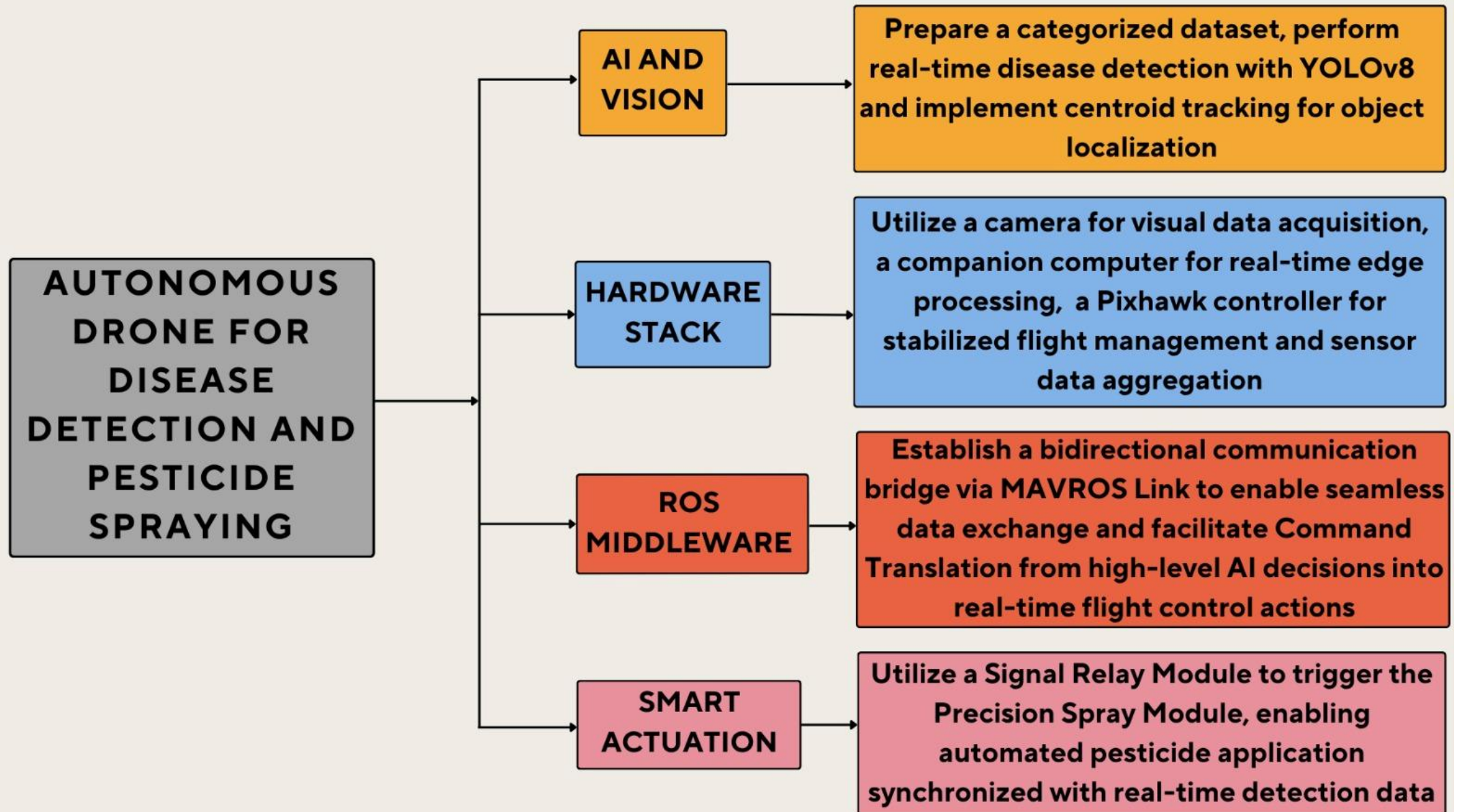
Research Gap

- Existing agricultural AI systems focus on leaf level disease detection and do not address disease identification on Arecanut bunches.
- Conventional agricultural drones rely on complete canopy spraying, which leads to inefficient pesticide usage and chemical wastage.
- Many existing systems depend on ground based processing, which results in high latency and reduced real time decision making.
- Most research is restricted to software level implementation, which consists of limited integration between AI models and real time hardware control

Objectives

- To collect the Arecanut clusters dataset and classify them into healthy and diseased (Koleroga) categories through annotation.
- To deploy a YOLOv8 based model for real time disease detection and perform target localization to generate precise coordinates for the drone's spraying mechanism.
- To design and assemble an autonomous quadcopter frame integrated with a vision system and tailored spraying payload for operation in dense Arecanut plantations.
- To implement a hardware integrated spraying mechanism triggered by real time disease detection.

Methodology



Methodology

Hardware Architecture

- A high performance companion computer runs heavy AI inference locally, eliminating ground station latency.
- The Pixhawk controller manages a stable navigation, GPS positioning, and sensor fusion.
- High resolution cameras serve as the primary data acquisition source.

ROS Middleware

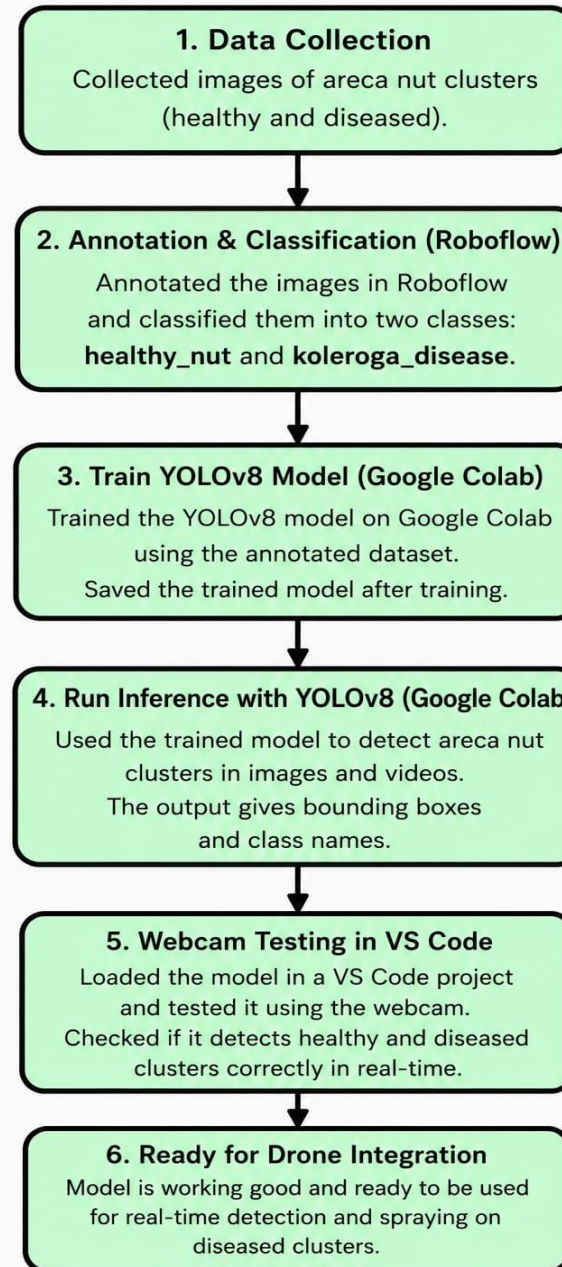
- Utilizes MAV ROS link to seamlessly connect the AI vision system with the physical flight hardware.
- Converts high level detection decisions into a low level MAV LINK protocols to instantly adjust flight behaviour and trigger the actuators.

Smart actuation

- Signal relay module acts as an electronic switch, instantly engaging the pump upon receiving a “Detect” command from the AI
- The precision spray module applies variable rate pesticide strictly to localized disease centroids, maximizing crop protection and minimizing chemical runoff.

Methodology

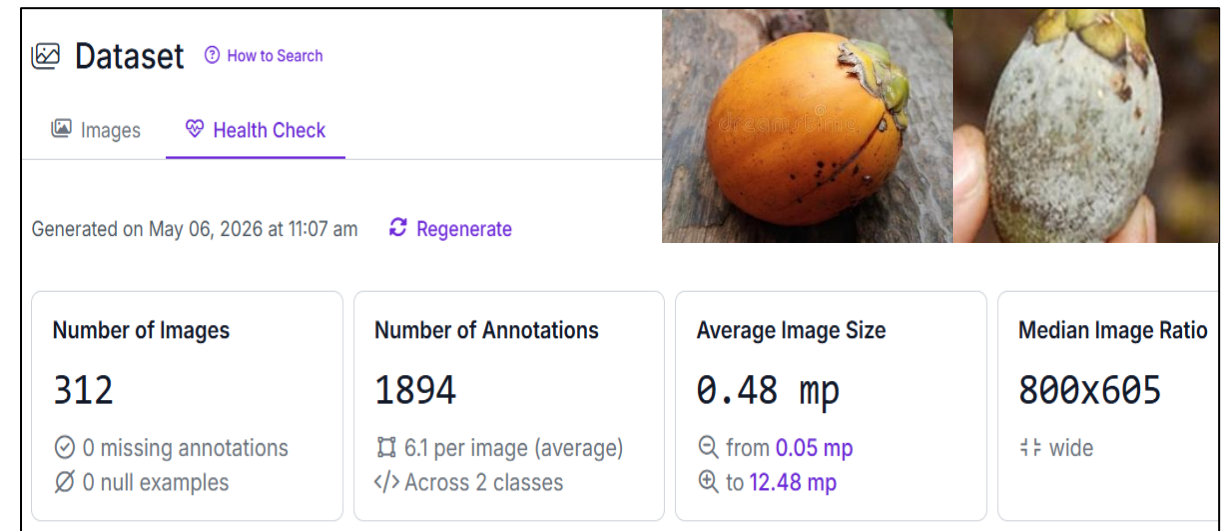
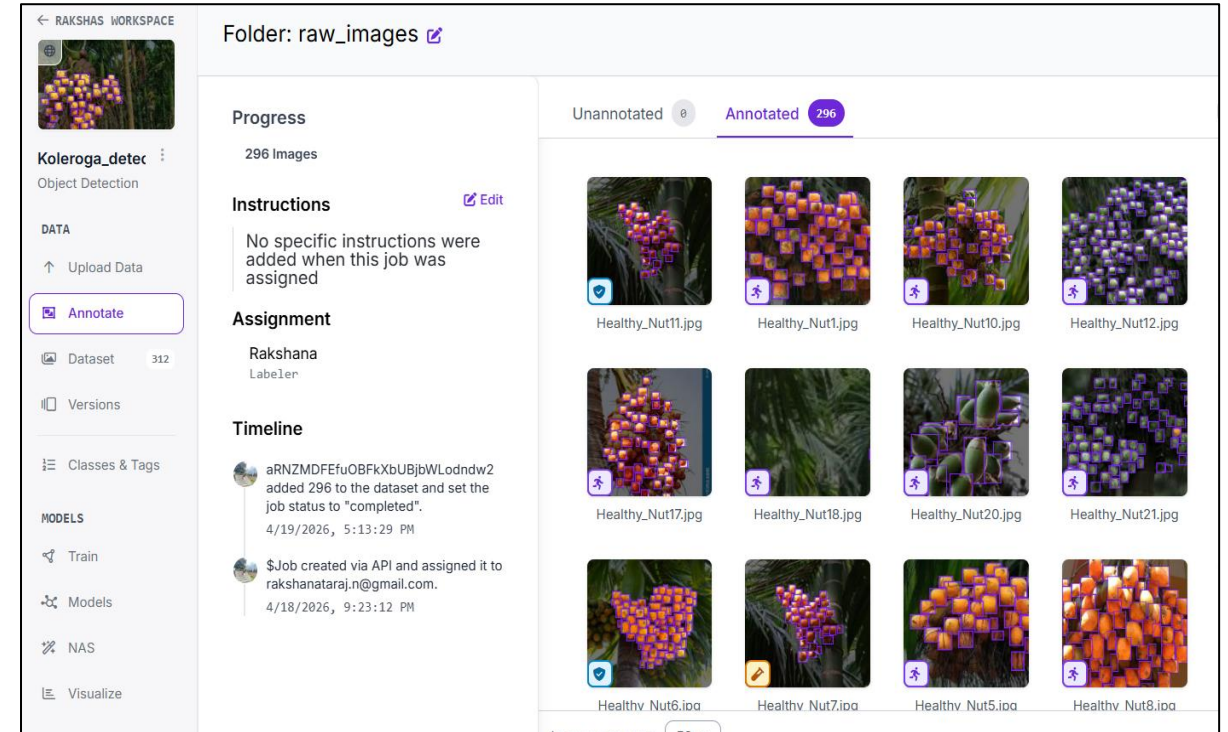
Software Architecture



Methodology

1) Data Sourcing and Annotation

- Collected a dataset of 300+ images consisting of Healthy and Koleroga infected nuts.
- Image annotation were performed manually using Roboflow for labeling the healthy and diseased nut.
- Defined two classes for classification: Healthy_nut and Koleroga_disease
- Organized and preprocessed the dataset for efficient YOLOv8 model training.



Methodology

2) Model Training using Google Colab

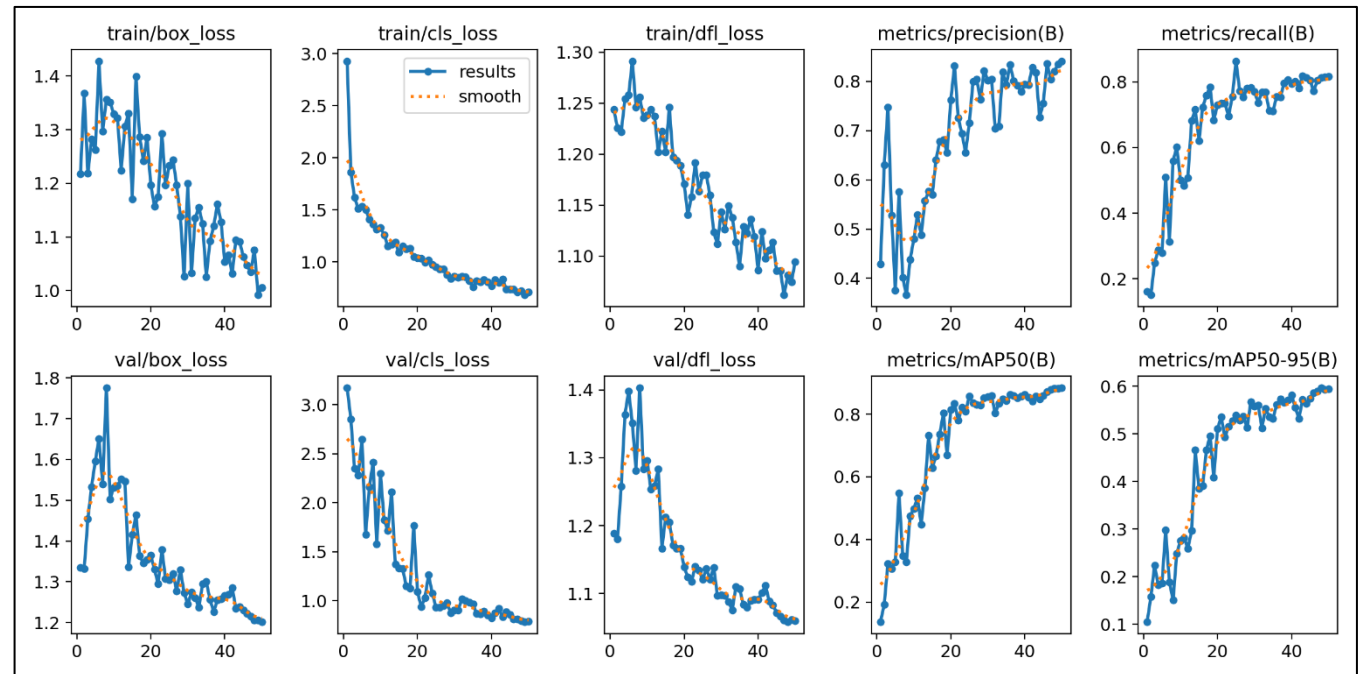
- Model training was performed on cloud based NVIDIA Tesla T4 GPU to handle deep learning computation.
- Utilized the YOLOv8 nano model for a lightweight and fast real time detection.
- Trained model for 50 Epochs using 512 x 512 image resolution for good detection accuracy.
- Successfully compiled and exported the best.pt weights file, containing the complete trained neural network

```
from ultralytics import YOLO

# Load the blank 'nano' brain
model = YOLO('yolov8n.pt')

# Start the training!
results = model.train(data=f"{dataset.location}/data.yaml", epochs=50, imgsz=512)
```

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	16/16	4.3it/s	3.7s
42/50	2.25G	1.031	0.8038	1.098	16	512: 100%	16/16	4.3it/s	3.7s
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%	1/1	3.0it/s 0.3s
	all	31	182	0.828	0.781	0.841	0.532		
43/50	2.25G	1.095	0.8359	1.106	16	512: 100%	16/16	6.1it/s	2.6s
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%	1/1	3.4it/s 0.3s
	all	31	182	0.818	0.819	0.857	0.572		
44/50	2.25G	1.092	0.7373	1.114	98	512: 100%	16/16	5.9it/s	2.7s
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%	1/1	4.3it/s 0.2s
	all	31	182	0.727	0.813	0.849	0.564		
45/50	2.25G	1.064	0.7374	1.086	20	512: 100%	16/16	6.0it/s	2.7s
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%	1/1	4.2it/s 0.2s
	all	31	182	0.756	0.804	0.86	0.574		

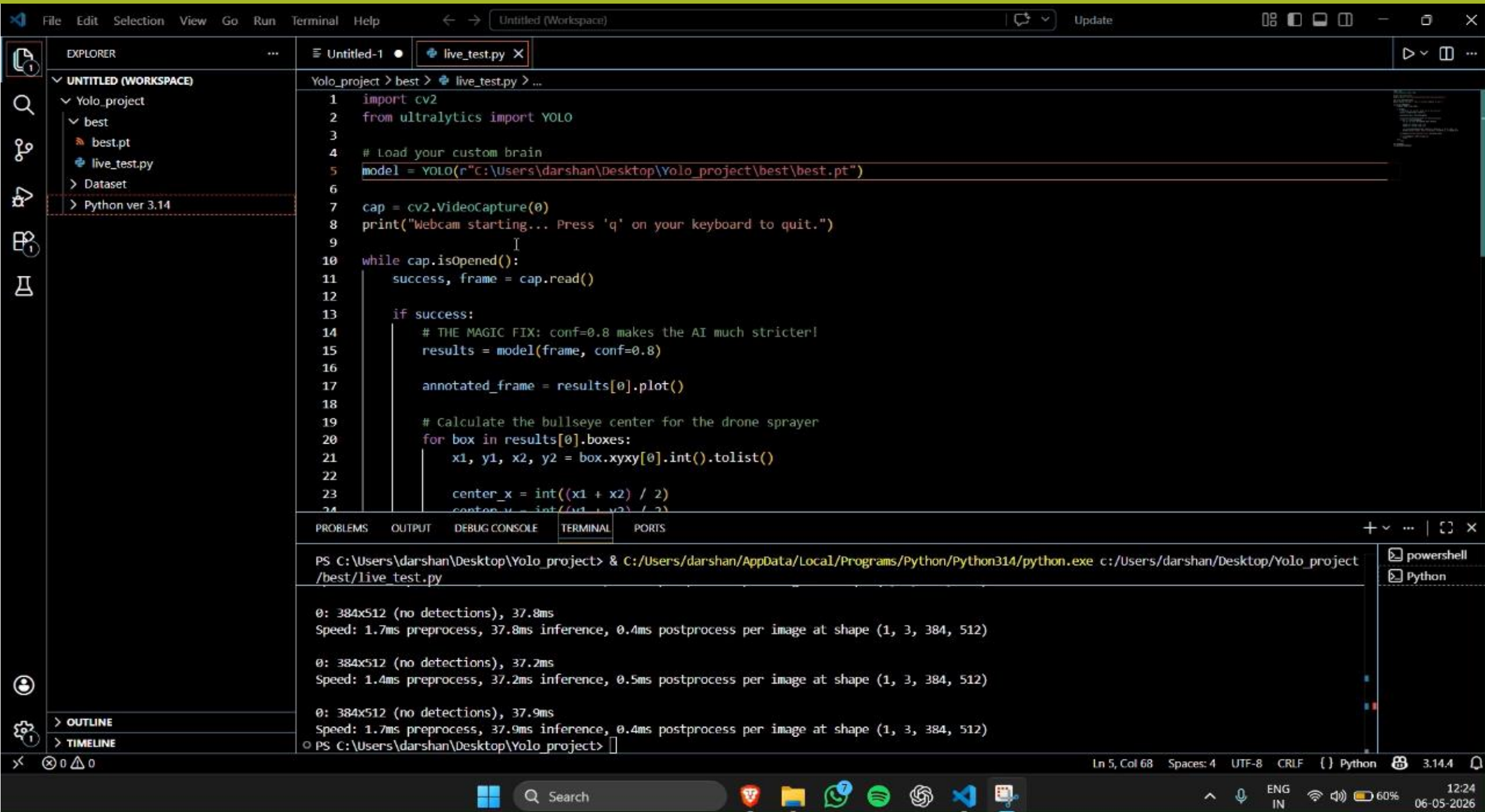


Methodology

3) Real time detection and targeting

- Deployed the trained YOLOv8 model locally using the visual studio code for real time disease detection.
- Implemented a 80% confidence threshold to reduce false detections in plantation environments.
- Developed centroid targeting logic to calculate the centre coordinates of detected diseased regions for precise spray targeting.
- Validated live inference using webcam feed for real time detection and tracking of infected arecanut clusters.

```
Untitled-1 • live_test.py X
Yolo_project > best > live_test.py > ...
1 import cv2
2 from ultralytics import YOLO
3
4 # Load your custom brain
5 model = YOLO(r"C:\Users\darshan\Desktop\Yolo_project\best\best.pt")
6
7 cap = cv2.VideoCapture(0)
8 print("Webcam starting... Press 'q' on your keyboard to quit.")
9
10 while cap.isOpened():
11     success, frame = cap.read()
12
13     if success:
14         # THE MAGIC FIX: conf=0.8 makes the AI much stricter!
15         results = model(frame, conf=0.8)
16
17         annotated_frame = results[0].plot()
18
19         # Calculate the bullseye center for the drone sprayer
20         for box in results[0].boxes:
21             x1, y1, x2, y2 = box.xyxy[0].tolist()
22
23             center_x = int((x1 + x2) / 2)
24             center_y = int((y1 + y2) / 2)
25
26             cv2.circle(annotated_frame, (center_x, center_y), 5, (0, 0, 255), -1)
27             print(f"Target Locked! Center coordinates: X={center_x}, Y={center_y}")
28
29             cv2.imshow("Koleroga Detection Live", annotated_frame)
30
31             if cv2.waitKey(1) & 0xFF == ord("q"):
32                 break
33         else:
34             break
35
36 cap.release()
37 cv2.destroyAllWindows()
```



CONCLUSION

The project successfully trained and deployed a custom YOLOV8 model capable of detecting Koleroga disease in real time, while being optimized for edge computing devices. An OpenCV based targeting system was also developed to accurately calculate the X and Y centroid coordinates, allowing the system to focus on an exact target point instead of just detecting a bounding box. The complete detection was tested and verified using both static images and live camera feeds, where the system effectively filtered background noise with more than 80% confidence accuracy. The software architecture is now fully verified and ready for integration with the Raspberry Pi companion computer and Pixhawk flight controller for autonomous payload testing.

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